

Homework 4
COMS 3310
Due Thursday April 2nd
or Friday 3rd for 15% Late Penalty

Please read the **HW guidelines** (see next page). **Failure to follow the guidelines will result in a lower grade.**

1. Design a CFG for each of the following languages using JFLAP, using the indicated file-names, and submit them all to **Homework 4 JFFs**. This part of the Gradescope assignment will be available latest by Thursday evening. (8 points each). Please click on <https://www.jflap.org/tutorial/> and click on **Grammar** for guidance on how to type in a CFG.
 - (a) The language L_1 of strings over $\Sigma = \{a, b, c\}$ in which every b is preceded and succeeded by an a (there is an a immediately before and immediately after every b). This means the immediate surroundings of every b looks like aba . Note that $ababa$ is in this language. Also, note that this is a regular language, so you may find it helpful to design a DFA first before designing the CFG from the DFA. Submit this as **hw4q1a.jff**.
 - (b) $L_2 = \{a^i b^j c^k \mid 0 \leq i \leq j \leq 2i \text{ and } k \geq 2\}$, as **hw4q1b.jff**.
 - (c) $L_3 = \{a^i b^j c^k \mid k = i + j\}$, as **hw4q1c.jff**.
 - (d) Let L_4 be a language over the alphabet $\{a, b, c, d\}$, defined inductively as follows (incidentally, this is a language of balanced parentheses, in which a and b are open and close parentheses of the same type, and also, c and d are matching parentheses of a different type). Submit as **hw4q1d.jff**. Ideally we would want to use $($ and $)$ instead of a and b , and also use $[$ and $]$ instead of c and d , but I'm playing it safe in case JFLAP doesn't like those special characters.
 - $\epsilon \in L$.
 - If $x \in L$, then axb and $cx d$ are in L .
 - If x and y are in L , then xy is in L
 - (e) The language L_5 of palindromes over $\Sigma = \{a, b\}$ which contain at least four b s. Submit as **hw4q1e.jff**.
2. Use the Pumping Lemma to prove that the following languages are not regular. (40 Points)
For this problem and the next, submit the handwritten and scanned answers on Gradescope **Homework 4 PDFs**.

- $L_6 = \{0^i 1^j \mid j = i \times i\}$. Please start this on a new page.
 - $L_7 = \{(ab)^i c^k \mid i = k \text{ and } i \geq 0\}$. This includes strings like $ababcc$ and $abababccc$. Please start this subquestion on a new page.
3. (20 points) Read the class notes (the last section) at the end of **CFG.pdf** which was published the week before Spring break. Then show that the following language is a CFL. Give a formal proof with the same level of rigor as in the notes. The proof here should be easier, but the approach required is the same. You need to show that a string is in L_8 if and only if it is generated by a CFG (which you will provide).

$$L_8 = \{a^i b^2 a^{2i} \mid i \geq 0\}$$

Again, please start this on a new page.

Guidelines:

- Please submit your HW via Gradescope.
- You must handwrite your solutions.
- If your handwriting is not legible, it will not be graded.
- Start answer to each question on a separate page.
- You are allowed to collaborate with *others*, while doing your homework.
- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one week of returning the homework. No requests will be entertained after this one week period.